

sequence number is called ISS (initial send sequence) for the receiver. If the random number is 250, this value is assigned to the calling sequence. If, and when, the receiver receives this pack correctly, it responds by sending a pack to the sender with both SYN and ACK flags set as 1 (SYN = 1 and ACK = 1).

The receiver chooses a sequence number by its random generator and sends a response message with this sequence number. If the chosen number is 350 (IRS, or initial receive sequence, number for the sender), the reply message will contain 350 in the sequence number field. The acknowledge number field contains the value 251 (meaning, the receiver is awaiting a message with sequence number 251 as it has already received message with sequence number of 250).

Once the sender receives the correct pack from the receiver, the former can send another pack with ACK flag set to 1 (that is, ACK = 1), with sequence number field set to 251 and acknowledge number field set to 351. The connection is then established.

After the connection is established, communicating parties can perform transmission in full-duplex mode. Connection can be terminated by any station, either sender or receiver, by sending a pack with FIN flag set (that is, FIN = 1). For example, after transmitting all data, the sender can send a pack with FIN flag set to 1. When this pack is acknowledged by the receiver, connection is terminated.

During connection request phase, the initial sequence number should be so chosen that previous connections (sockets) are not confused with new connection requests (sockets). This typically happens when a host application crashes and, before the other side times out, re-establishes the crashed connection quickly by recognising that the connection request is for the old socket.

ISN selection is made by a 32-bit random generator and is created during connection request phase. The number is generated by a 32-bit clock, which has ISN cycle of 4.55 hours. The clock is incremented approximately every 4μs, for which a unique

Source Port (16 bits)	Destination Port (16 bits)
Length (16 bits)	Checksum (16 bits)
Data (Variable)	

Fig. 2: UDP headers

ISN is generated with a cycle period of 4.55 hours.

- A 16-bit window flag, which indicates the number of bytes that the sender can transmit without receiving acknowledgements from the receiver. This field is used by the receiver to inform the sender about the availability of the buffer in the receiver. Based on this information, the sender controls the flow. Flow control is implemented by the window field.

- The urgent pointer field, which is used to alert the receiver about incoming urgent data and indicates the end of the urgent data within the sequence of the transmission of packs. Value in the urgent pointer is valid, provided URG (urgent) flag is set (that is, URG = 1). Urgent pointer value defines the end of urgent data and the start of normal data in bytes. For example, if urgent pointer is 0000000001000000, the receiver will understand that the next 64 bytes of data will be urgent.

- Reserved bits (six bits) are for future use. All flags—URG, ACK, SYN, FIN, RST and PSH—together are known as control fields of TCP. PSH flag is set when the receiver is told to pass the data immediately to the application. Otherwise, the receiver will buffer the segments until sufficient storage is made. RST flag is used to abort TCP connection. When this flag is set, the receiver is advised to terminate the connection for an abnormal condition.

The options header can be of maximum 40 bytes and is always a multiple of four bytes. To make it, a multiple of four bytes padding may be used, if required. A few important options of TCP pack are:

Maximum segment size (MSS) option. This is used to indicate the maximum size of the segment that the sender can accept. This option is used during connection establishment phase. The sender specifies the MSS. This is a 16-bit option field. The largest

block that MSS can specify is $2^{16}-1 = 65,535$ bytes. A TCP pack or segment has minimum 20-byte header like that of an IP header. Thus, the largest block size of data that a TCP segment or IP datagram can carry is of $65,535 - 40$ bytes = 65,495 bytes. Declaration of MSS during connection setup is made to overrule TCP default MSS. TCP default MSS is 536 bytes.

Window scale option (WSO). The TCP pack has a header of 16-bit window. Maximum permissible advertised window size is $2^{16}-1 = 65,535$ bytes. With WSO, the larger advertised window size may be used for which upward scaling up to 2^{14} is permissible. With WSO, scaled advertised window size may, hence, go up to $2^{14} \times 2^{16}-1 = 1.073,725,440$ bytes.

Times stamp option. This is used for calculating round trip delay, which is used for calculating time-out period useful for flow control.

TCP divides data into manageable packets that are easier to deal with. It provides a guaranteed error-free connection. TCP breaks a network terminal (NT) service pack into several packets. If an error occurs, a small packet can be retransmitted instead of the whole pack.

TCP uses a header to establish a virtual connection to ensure successful transmission of packets and completion of transmission of the whole message. Source and destination port fields keep the trace of packets. Packets belonging to various applications, such as e-mails, on-line charts and multiple running browsers, can run concurrently based on the trace of packets. Sharing of an established virtual link is allowed at the same time. Sequence number is used to reassemble the packets at the receiver possibly received out of sequence due to sharing.

The receiver, on receiving a packet, performs check-sum operation to detect errors. If the packet is received error-free, a positive acknowledgement is sent. If packet is received erroneously, a negative acknowledgement is sent.

The transmitter, on getting either a negative acknowledgement within a time known as time-out period (TOP) or not getting any response within